EMPIRICAL HYDROLOGICAL, TOPOGRAPHICAL & STRUCTURAL DATA REGISTRY

Nifas Silk-Lafto Sub-City Drainage Network Research Dataset (Addis Ababa, Ethiopia)

Target Corridor: Jemo 1 → Jemo 2/3 → German Sq. → Lancha
Standards: ERA Drainage Manual & AACRA Standards
Terrain Model: Copernicus / OpenTopography 30m DEM
Document Scope: Full Research Baseline Empirical Dataset

1. TOPOGRAPHICAL & SUB-CATCHMENT BOUNDARY DATA

Topographical elevations, bed slopes, and catchment boundaries extracted from OpenTopography / Copernicus 30m DEM across the Jemo–Lafto drainage basin:

Parameter ID	Node 1 (Jemo 1)	Node 2 (Jemo 2/3)	Node 3 (German Sq.)	Node 4 (Lancha Outlet)	Source / Standard
Coordinates	8.9482° N, 38.7105° E	8.9391° N, 38.7182° E	8.9284° N, 38.7290° E	8.9195° N, 38.7412° E	OpenStreetMap / GIS Trace
Upstream Elev.	2,240 m.a.s.l.	2,210 m.a.s.l.	2,192 m.a.s.l.	2,183 m.a.s.l.	DEM Elevation Extract
Downstream Elev.	2,210 m.a.s.l.	2,192 m.a.s.l.	2,183 m.a.s.l.	2,172 m.a.s.l.	DEM Elevation Extract
Length (L)	2,500 m	2,250 m	2,000 m	2,000 m	Direct Channel Reach
Bed Slope (S)	**0.0120 (1.20%)**	**0.0080 (0.80%)**	**0.0045 (0.45%)**	**0.0055 (0.55%)**	(E_start - E_end) / L
Area (A)	**48.0 ha**	**65.0 ha**	**42.0 ha**	**58.0 ha**	Hydro-GIS Delineation
Runoff Coeff (C)	**0.80**	**0.85**	**0.82**	**0.78**	ERA Manual Table 5-2

2. RAINFALL IDF EMPIRICAL PARAMETERS (ADDIS ABABA STATION)

Derived from the ERA Drainage Design Manual (Region A2: Central Ethiopian Highlands) using the empirical IDF formula:

$$I = (a \cdot T_r^{m}) / (t_d + b)^{c} \quad | \quad a = 520.4, \;\; b = 14.5, \;\; c = 0.78, \;\; m = 0.19$$

Design rainfall intensities for short-duration flash storms (t_d = 30 minutes):

Return Period (T_r)	2-Year	5-Year	10-Year	25-Year	50-Year	100-Year
Rainfall Intensity (I)	**42.50 mm/hr**	**54.80 mm/hr**	**68.00 mm/hr**	**81.40 mm/hr**	**94.20 mm/hr**	**108.50 mm/hr**

3. AACRA STRUCTURAL CONDUIT & CULVERT SPECIFICATIONS

Reinforced concrete box culvert dimensions as per Addis Ababa City Roads Authority (AACRA) design standards:

Parameter ID	Node 1	Node 2	Node 3	Node 4	Standard Source
Culvert Class	Single Box Culvert	Twin/Wide Box Culvert	Single Box Culvert	Double Trunk Box	AACRA Standard Drawings
Inner Width (B)	**1.50 m**	**1.80 m**	**1.50 m**	**2.00 m**	Physical Sizing Target
Inner Height (H)	**1.20 m**	**1.40 m**	**1.20 m**	**1.50 m**	Physical Sizing Target
Manning's n	**0.015**	**0.015**	**0.015**	**0.015**	Concrete with sediment
Clean Capacity Q_{cap}	**5.12 m³/s**	**7.04 m³/s**	**3.14 m³/s**	**6.28 m³/s**	Manning Eq. (β = 0.0)

4. MAINTENANCE SILTATION & CAPACITY REDUCTION MATRIX

Effective hydraulic capacity Q_{cap} calculated across siltation levels ($\beta \in [0.0, 0.75]$) using Manning's Equation:

$$H_{eff} = H \cdot (1 - \beta) \quad | \quad A_{eff} = B \cdot H_{eff} \quad | \quad Q_{cap} = (1/n) \cdot A_{eff} \cdot (A_{eff} / (B + 2H_{eff}))^{2/3} \cdot S^{1/2}$$

Blockage Ratio (β)	Clogging Description	Node 1 Q_{cap}	Node 2 Q_{cap}	Node 3 Q_{cap}	Node 4 Q_{cap}
β = 0.00	Fully Clean / Desilted	5.12 m³/s	7.04 m³/s	3.14 m³/s	6.28 m³/s
β = 0.15	Minor Siltation (15%)	4.28 m³/s	5.90 m³/s	2.63 m³/s	5.26 m³/s
β = 0.30	Moderate Blockage (30%)	3.41 m³/s	4.72 m³/s	2.10 m³/s	4.21 m³/s
β = 0.50	Severe Clogging (50%)	2.23 m³/s	3.11 m³/s	1.37 m³/s	2.75 m³/s
β = 0.75	Critical Overflow (75%)	0.85 m³/s	1.20 m³/s	0.52 m³/s	1.05 m³/s

5. BASELINE AUDIT MATRIX: 10-YEAR STORM (68.0 MM/HR) & B = 0.30

Node ID	Corridor Location	Q_{peak}	$Q_{capacity}$	Capacity Factor (CF)	Audit Status	1-Hr Surcharge Vol
Node 1	Jemo 1 Upper	7.25 m³/s	3.41 m³/s	**2.13**	CRITICAL OVERFLOW	13,824 m³
Node 2	Jemo 2/3 Middle	10.44 m³/s	4.72 m³/s	**2.21**	CRITICAL OVERFLOW	20,592 m³
Node 3	German Sq. Flat	6.51 m³/s	2.10 m³/s	**3.10**	CRITICAL OVERFLOW	15,876 m³
Node 4	Lancha Outlet	8.55 m³/s	4.21 m³/s	**2.03**	CRITICAL OVERFLOW	15,624 m³
TOTAL	**Corridor Summary**	**32.75 m³/s**	**14.44 m³/s**	**2.27 (Avg)**	SYSTEM FAILURE	**65,916 m³**